

Defense Brief — Representation Convergence Project

Companion to the 38-page report. One page of framing per topic; every number cited is measured and appears in the report with its generator notebook.

1 · Assumptions (what the results are conditional on)

Item-level correspondence exists and is known for supervised fits. Row i of every array is the same passage or image in every model. This is constructed (nested sorted-id prefixes, deterministic caption recipe) and verified numerically wherever two caches meet; C.12 is the one experiment that deliberately removes it.

Linear maps, closed form. Every fitted map is ridge regression solved in closed form - no gradients, no iteration - so 'related by a linear transformation' carries no asterisk. The MLP contrast (F2) is the deliberate exception and converged to the linear answer exactly.

Frozen encoders. No pretrained weight is modified anywhere in the evidence chain; F3, which would unfreeze one, is built but gated shut to preserve that statement.

Held-out discipline includes the transforms. Splits precede fitting; whitening covariances, hub bases, and alpha choices come from training rows only. Gallery size is 1,000 everywhere, so chance is 0.001 and R@K columns are comparable across experiments.

Geometry diagnostics are trusted as measurements. Effective rank (spectral entropy) and mean pair-cosine are treated as valid summaries of a space's shape; C.14's perfect rank ordering is itself evidence the trust is earned.

2 · Restrictions (scope the claims do not exceed)

claim	measured scope	explicitly NOT claimed
convergence is linear	GPT-2/Pythia; MobileCLIP/SigLIP; DINOv2 x3; 4 text encoders	all architectures, all scales
alignment scales with capacity	DINOv2 family, 22M-304M (13.8x), compressive at the top	behaviour beyond 304M; other families
isotropy rescue	GPT-2 target, two image scales (102.6%/101.4%)	sufficiency - C.14 bounds it
unsupervised recovery	39.7% exact @ 595x chance; 40% of ceiling	a deployable pairing-free system
shared hub	7 encoders, 4 objectives, COCO, English; transfer 93-96%, cross-lineage 94.2%	a universal space for ALL models or architectures
writing INTO a space	hub reads OUT of every encoder; writes back into all but collapsed GPT-2 (24.8%)	that a collapsed space can be reconstructed for ranking
structure is shared	LOCALLY - kNN ratio 100s x chance; falls to ~2x by $k=500$	global metric identity - it diverges at scale
deployment numbers	COCO val/train2017 domain; 1,000-item galleries	open-world scale or other domains

One language, one image domain, one caption corpus throughout. Every 'per cent of ceiling' is against a ceiling re-measured under the same protocol and caption treatment - never a borrowed number.

3 · Tuning (every knob, and how it was set)

knob	how set	guard against overfitting the knob
ridge alpha	validation sweep $1e-3..10$ per experiment	chosen on train-internal validation, never on eval
pooling	fixed cls+patch across the scale sweep	the one mixed run (base CLS) excluded from comparisons
N (pairs)	scaled with encoder width -> 11.1 rows/dim at every sweep point	starvation biases DOWN, so it cannot fake the trend
GW epsilon	swept per run; optimum MOVES with N ($5e-3@800$ -> $1e-3@1500$)	reported as a sweep, never a single tuned value
HUB_DIM x alpha	2-D sweep, transfer-first, rows/dim ≥ 5 enforced	pre-registered thresholds; cliff reported as finding
gates	90%-of-ceiling; 85% transitivity; 50% transfer; ≥ 5 rows/dim	registered before results; 89.0% reported as NOT met

4 · Defense — the hard questions, with answers

“Isn't whitening cheating?” It is label-free (uses no pairs, no labels), fitted on training targets only, and linear - so it is an instance of the thesis, not an exception. The full map is whitened composed with ridge: still one matrix.

“Why is the whitened R-squared 0.014?” Whitening destroys magnitudes and perfects angles. R-squared asks about magnitudes; retrieval asks about angles. One system, two metrics, opposite verdicts - which is the project's metric-geometry lesson in its purest form.

“You said isotropy helps - then why does it destroy C.12?” Reading versus finding. Cosine reads a KNOWN correspondence: uneven variance is noise. Gromov-Wasserstein FINDS an unknown one from distance structure: uneven variance is the signal that makes points distinctive. Same property, opposite prescriptions - both measured, twice each.

“Doesn't C.14's residual spread undermine C.11?” It bounds it. C.11's mechanism predicted two unseen encoders' ordering perfectly (rank-corr +1.00). The residue after whitening (0.110) measures what contrastive training adds BEYOND the frame: the arrangement of meaning across directions. Necessary, not sufficient - a stronger theory than the original.

“Why did the hub break at 1024 dimensions?” Whitened PCA divides by singular value; past ~512 it amplifies noise-dominated tail directions, which take over cosine geometry. Not a power artifact (rows/dim ≥ 5 on both sides of the cliff); alpha shifts it but cannot substitute for width. The cliff is the third measured instance of portability-versus-losslessness.

“Is the transfer result circular - the head just learned the task?” The random-map control replaces only the hub map and collapses to chance (0.001), so the head alone carries nothing. And absolute R@1 RISES on unseen encoders, which a task-memorising head cannot produce.

“The hub fails to write into GPT-2 - isn't that a hole?” It is a bounded and explained one. The hub READS every encoder out at 91-99%, including GPT-2 at 95.1%; it cannot WRITE back into GPT-2's space (24.8%). The cause was tested, not assumed: the hub reconstructs that space essentially perfectly, so nothing is lost - the failure is that ranking WITHIN a space whose items sit at +0.999 cosine from each other needs precision finer than one extra fitting stage preserves. For the deployed use, encoders meet IN the hub and GPT-2 reaches the bge readout at 93.9%, so the limitation never binds. It is a property of the target's geometry, not of the method. (C.13.3-4.)

“Is the shared structure real or just local?” Local, and measured to be so. Neighbourhood overlap runs hundreds of times chance at $k=1$ and falls to roughly 2x by $k=500$ for all 21 pairs - the encoders agree strongly on what is NEAR what and diverge about global arrangement, exactly as two map projections preserve local adjacency while distorting global distance. This is stated as a restriction, and it also explains why a linear adapter reaches most but not all of the ceiling: the part it misses is local structure no single linear map can carry. (C.13.5.)

“Doesn't the cross-lineage test still share an objective?” Yes, and the brief says so. SigLIP 2 transfers at 94.2% with no DINOv2 ancestry - inside the within-family band, so shared lineage was not doing the work. But SigLIP is contrastively trained on image-text, so the result is cross-lineage, NOT simultaneously cross-lineage and cross-objective, and it is still a Vision Transformer. A convolutional encoder is the stronger open test, and the notebook accepts one as a one-line change. (C.13.2.)

“Why is bge-m3 the readout target if SBERT scores higher?” Because every ceiling and every percentage-of-ceiling figure in the project is measured against bge; switching in the final sections would make the numbers incomparable for a gain of about three points. SBERT is genuinely better as a target (101.0% writable against 98.1, raw R@1 0.510 against 0.466) and is offered as an alternative in the demo. bge is retained for comparability, not because the data chose it - stated plainly rather than presented as a merit ranking. (C.13.4.)

“How do you defend the interpretations against wishful reading?” Several explanations offered during the work were TESTED and FAILED, and are recorded as failed rather than quietly dropped: that collapse would depress shape correlation (false - rank measures survive it), that the hub discards a collapsed space's directions (false - it reconstructs them perfectly), that reach-into-hub predicts writability (false - effective rank does), and that adding encoders improved transfer (false - the gain was six queries in a thousand, within noise). A claim that survived a test that could have killed it is worth more than one that was never exposed to one.

“How far does the shared hub actually go?” Seven encoders, four text training objectives, two modalities, one 512-d linear coordinate system. Composition through it retains 95.6 per cent of directly-fitted pairwise maps across 36 of 41 ordered pairs, and adding encoders IMPROVED transfer (R@1 0.451 at seven spaces against 0.442 at five). The five exceptions share one target: writing into GPT-2's collapsed space, at 24.8 per cent. Reading OUT of it works at 95.1.

The cause was tested, not assumed - the obvious explanation, that the hub discards that space's directions, is false; the hub reconstructs it perfectly, and the failure comes from ranking within a space whose own items sit at +0.999 cosine from each other. (Appendix C.13.3.)

“Could the agreement be an artefact of the fitting?” No, and it is measurable without any fit. Comparing each space only to ITSELF - how far apart are items i and j inside it - and correlating those answers across spaces needs no map, no alignment and no labels. Cross-modal pairs reach rho +0.29 and 67 per cent nearest-neighbour overlap at 27 times chance, while cross-space cosine sits at zero. The agreement cannot have been created by a fit that has not happened yet. Notably bge-m3 and SBERT reach +0.76, the highest of all 21 pairs, sharing no lineage - only a training objective. (Appendix C.11.2.)

“Have the artifacts themselves been checked?” Yes, in one automated run after all experiments were complete: 47 checks across every stored artifact - statistical power, row alignment by shuffle test, sanity, target geometry, and cache-key integrity. NO hard failures. Every fit sits between 5.9 and 9.8 rows per input dimension; shuffle gaps run 0.545 to 0.863 against a 0.2 threshold; all three encoder caches match the configuration named in their filenames. The only two flags are the GPT-2 caption space, whose collapse is a documented finding rather than a defect - and the battery reproduced its signature independently in two separately generated files. (Appendix D.5.)

“How do I know the numbers are right?” Five documented self-corrections via measurement (starvation rerun, synthetic-figure retraction, matched ceiling, wrong reference arm twice - now derived from filenames, whitening-for-GW reversal), plus width-tagged checkpoints, transfer checks against known pairs, and AST-validated notebooks. The error CLASS is real; the defenses are structural.

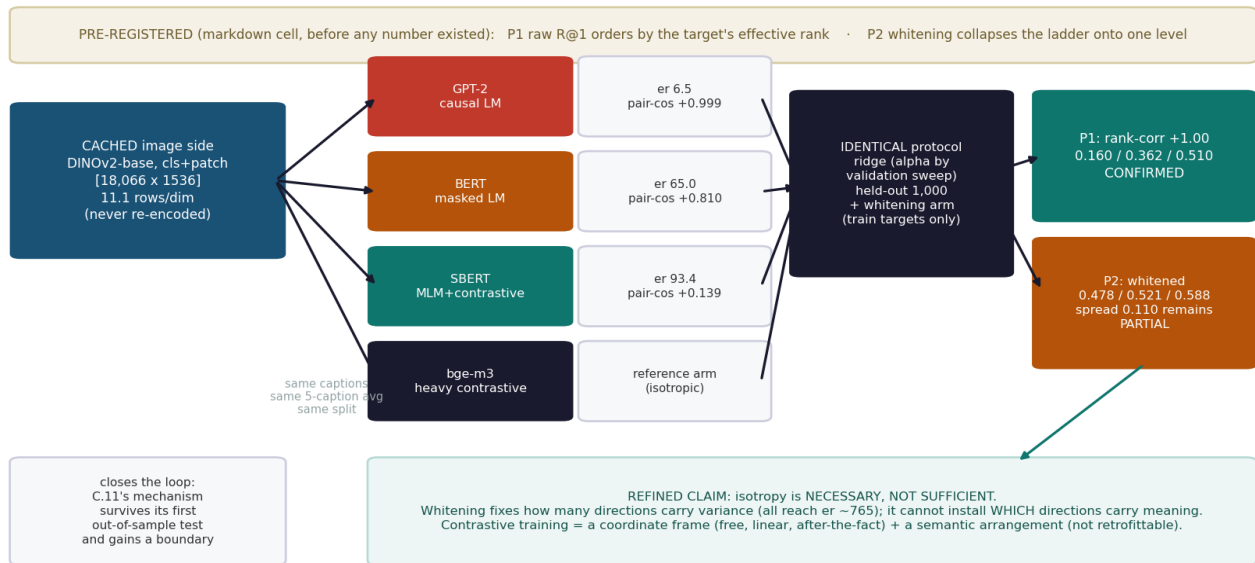
5 · Achievements — the nine findings

#	finding	headline evidence	section
1	convergence is linear	MLP = ridge exactly at 74 rows/dim (0.9052 = 0.9052)	C.9/F
2	related, not rigid	Procrustes 0.038 vs ridge 0.592; 69x singular-value spread	4, B
3	deployable	95% of ceiling @ 0.79 MB; verif AUC 0.999 / rank 0.987 / membership 0.889	4, D, D.2
4	cross-modal, no joint training	40.9% of matched ceiling, 358x chance	C.8-C.9
5	alignment scales with capacity	+15.3 pts of ceiling per decade, 6 monotone measures, compressive	C.10
6	isotropy, not similarity training	whitening -> 102.6% / 101.4% of bge at two scales; rho +0.94	C.11
7	correspondence is self-identifying	GW 39.7% exact = 595x chance, 0 pairs; + 2 measured negatives	C.12
8	one shared space, used	hub transfer 95.9%/92.9% of native on unseen encoders; control at chance	C.13
9	the mechanism predicts	pre-registered ladder: rank-corr +1.00; isotropy necessary-not-sufficient	C.14

Plus the recurring meta-finding, measured three times: isotropy helps reading a correspondence and hurts finding or losslessly carrying one (C.11 vs C.12 vs the C.13 width cliff).

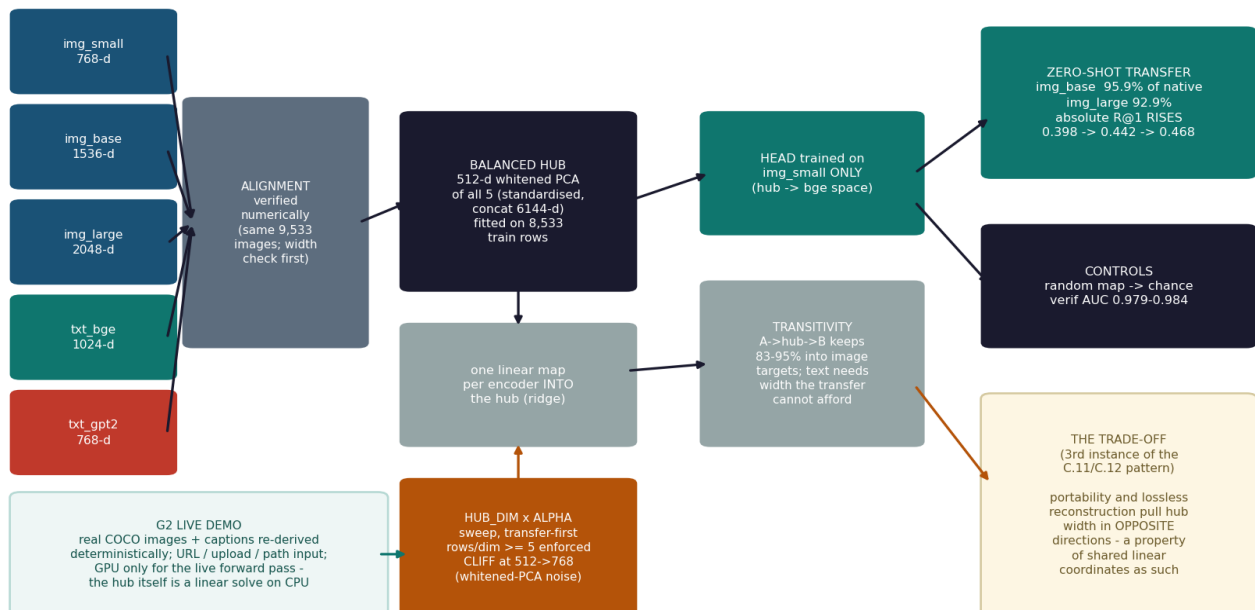
6 · Flow: how E1.3 closes the loop

E1.3 — the encoder ladder: one image side, four text objectives, predictions registered first



Flow: the hub, built and demonstrated

G1/G2 — five encoders, one 512-d hub: the shared space built, measured, and demonstrated live



7 · Conclusion

The thesis - independently trained models share representations up to a linear transformation - was tested in every posture available: pairwise and supervised (it holds, and an MLP adds exactly nothing); across modalities without joint training (it holds at 41-58 per cent of a jointly-trained ceiling, rising with capacity at a measured slope); with the metric geometry itself interrogated (apparent failures were coordinate-frame artifacts, removable by a free linear transform, with the residue bounded by a pre-registered prediction test); with the pairing itself withheld (the correspondence partially identifies itself from shape alone); and constructively (one 512-dimensional linear hub serves five encoders, and a component trained on one transfers to encoders it never saw at 93-96 per cent of native performance).

Two things distinguish the project beyond the results. First, its negatives are load-bearing and mechanistic: ICP converging on its own errors, whitening destroying discovery, the hub-width cliff - each measured, each explaining the boundary it marks. Second, its errors are part of the record: five self-corrections reached by measurement rather than argument, which is the difference between a demonstration and research. The final state of the claim is narrower and stronger than the initial one: the shared structure is real, linear, discoverable, and usable - and the coordinate frame that makes it readable is cheap, while the semantic arrangement that makes it meaningful is what training actually buys.